



As personal, corporate and even national computer systems get subjected to stealthy, continuous cracking processes, it makes sense to be warned and armed. Read on to find out more about one high profile case of a silent malicious cyber attack (APT) that brought a country's nuclear programme to its knees – Stuxnet.

**A**dvanced persistent threats or APTs are a kind of cyber attack that can go undetected for a long period of time. The objective of the attack is:

- To extricate vital data rather than cause damage.
- To gain access to the network or host.

The typical characteristics of APTs are listed below.

**Advanced:** They are highly sophisticated and targeted.

**Persistent:** They put in a lot of effort to achieve their goal and to reach the target.

**Threatening:** They have a newly generated payload compared to the conventional, target-specific payloads — something not known to security personnel.

In an APT, the attackers deface the network, plant advanced malware and hide themselves until they receive the command to get off the target. An advanced persistent threat uses a step-by-step approach.

### Intrusion kill chain

“Intrusion kill chain is crucial, but it's not enough.” -Peter Tran.

The intrusion kill chain (also known as a ‘kill chain’, a ‘cyber kill chain’ or ‘attack cycles’) is a model used to execute the successful pathway towards segmenting, analysing and mitigating the cyber attack. Various stages of attacks are performed in order to gain access to the target.

1. **Reconnaissance:** This is the initial stage of identifying and selecting the target. All the relevant information about the target is obtained by crawling the Internet (social engineering) to get personal information like email, social media ID, etc.
2. **Weaponisation:** At this stage, a RAT (remote administration tool or a Trojan) is encapsulated with any client application which is likely to be an executable.
3. **Delivery:** During this stage, the weaponised executable